

Thermodynamic Exergy Analysis of a Turboprop Engine—Part 2: Analysis, Comparison to Publication, Improvements, and Lessons Learned

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1. Introduction

In the first phase of the course project, focus was on conducting a literature review of exergy analyses of turboprop and gas turbine engines and the selection of a case study (from a publication) to perform an exergy analysis on. In this phase, the exergy analysis is performed on the case study and results are compared to the publication. Improvements to the turboprop engine and to the exergy analysis methodology are also provided along with lessons learned.

2. Case Study

The case study for this project is the T56 Turboprop engine analyzed in the journal article “Energetic and Exergetic Analysis of T56 Turboprop Engine” [1]. The article is included in Appendix VII (page 28). The T56 Turboprop engine is produced by Rolls-Royce (formerly Allison Engine Company) and is in widespread use around the world in both military and civilian aircraft applications [2]. It has a single shaft with a multi-stage axial flow compressor and a multi-stage turbine unit.

The main goal of this phase of the project is to reproduce results obtained in the article using appropriate simplifications to methodology where appropriate. The article’s authors used actual measurement data from a test cell in Turkey as the basis for their analysis. Data included measurements of pressure and temperature, torque and RPM, fuel flow rate, air flow rate, and residual thrust of exhaust gases. The test measured propeller shaft power without an installed propeller; hence, like in the article, my analysis does not reflect the propulsion efficiency of the propeller. Exergy analysis is performed on the individual turboprop engine components and the entire engine for the four power settings tested: 75%-mode, 100%-mode, MIL-mode, and takeoff-mode.

The publication describes two cases for entire-engine exergy performance evaluation: Case A, inclusion of shaft power and Case B, inclusion of both propeller shaft power and kinetic energy of exhaust gas. My focus is on Case B only, since this is a minor extension of Case A and is more relevant from a useful-power-output perspective.

My study results compare directly to the publication results shown in Table 1 and Fig. 2 below. Other exergy performance parameters, such as the publication fuel exergy depletion rate, productivity lack ratio, exergetic improvement potential, and exergetic fuel-production ratio, are not included in the analysis and comparison since they are just different ways of presenting exergy destruction.

Table 1

Exergy efficiency of the turboprop engine [1].

Parameter	Unit	Case	Operation Modes			
			75%	100%	MIL	Takeoff
Exergy efficiency	%	B	23.4	24.9	25.6	26.3

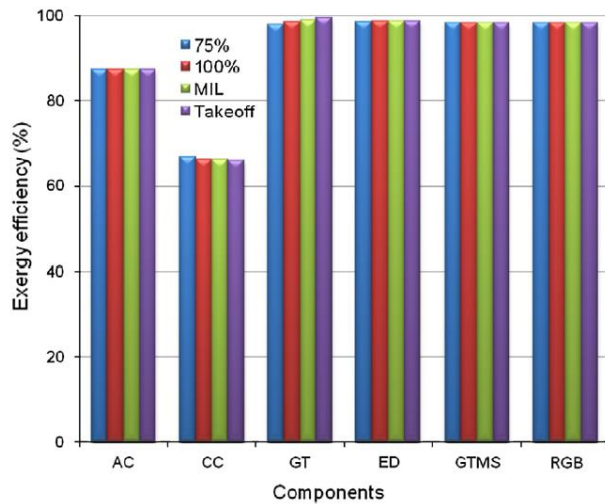


Fig 1. Exergy efficiency of the turboprop engine components for various operating modes [1].

3. Exergy Analysis Methodology

3.1 Overview

To conduct an exergy analysis, principles from both the first and second laws of thermodynamics are employed. A mass balance is first used to determine mass flow rates throughout the turboprop engine. First law energy balance is used to estimate the shaft power supplied to the compressor, the shaft power obtained from the turbine, the transmission of power along the power transmission system, as well as assess the efficiency of combustion. Second law methods are used to estimate exergy rates at each of the states throughout the engine and ultimately estimate the exergetic efficiency of each of the components and entire engine. State properties are determined at the entrance and exit of each of the engine components analyzed. Some of the properties are obtained by measurement data, others are calculated using first and second law methods. The entire turboprop engine and each of its components are assumed to be in steady-flow operation.

The main components of the engine are the air compressor (AC), combustion chamber (CC), gas turbine (GT), exhaust duct (ED), gas turbine mechanical system (GTMS), and gear box (GB). These components, and each of the required states are identified in Fig. 2.

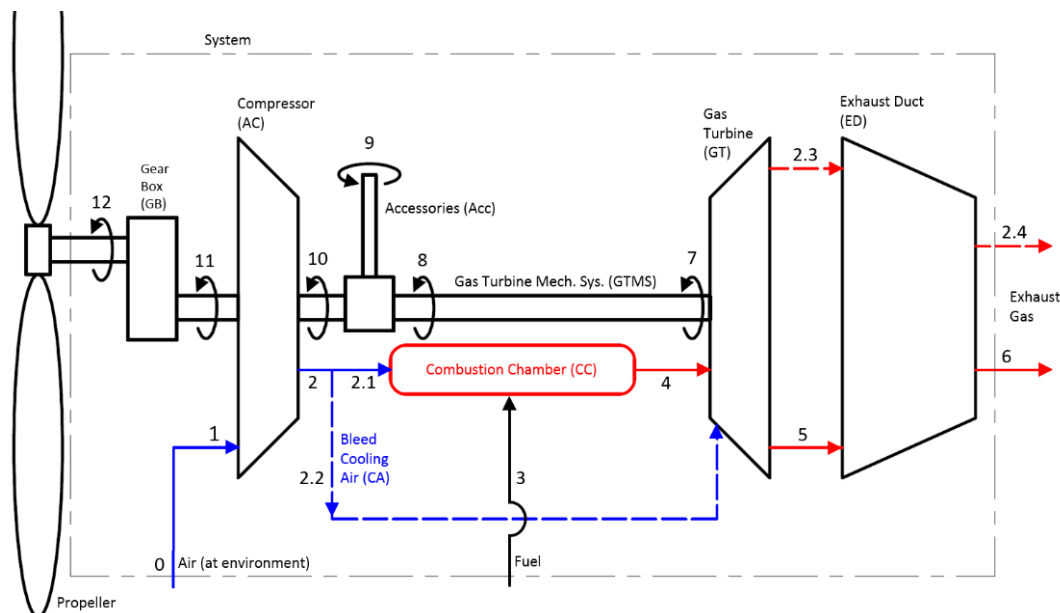


Fig 2. Basic components and state locations of the single-shaft turboprop engine.

3.2 First Law Analysis

Measurement data obtained from the turboprop engine test are included in Appendix II (page 13). The general mass balance (Eq. 1) was applied to the air and combustion gas pathways throughout the engine using the measured data:

$$\sum \dot{m}_{in} = \sum \dot{m}_{out} \quad (1)$$

$$\dot{m}_2 = \dot{m}_1 \quad (2)$$

$$\dot{m}_{2,1} = 0.9 \dot{m}_2 \quad (3)$$

$$\dot{m}_{2,2} = \dot{m}_{2,3} = \dot{m}_{2,4} = 0.1 \dot{m}_2 \quad (4)$$

$$\dot{m}_4 = \dot{m}_5 = \dot{m}_6 = \dot{m}_{2,1} + \dot{m}_3 \quad (5)$$

where $\dot{m}_{in}$ and $\dot{m}_{out}$ are the mass flow rates of the inlet and outlet streams of a component control volume, and $\dot{m}_{\#}$ are the mass flow rates of the states represented by the subscript placeholder #. It is assumed that 10% of the air flow rate at state 2 is redirected as cooling air for the turbine blades. Nomenclature is included in Appendix I (page 12). The first law energy balance for a component control volume, assuming steady-flow, is

$$\dot{Q}_{in,net} - \dot{W}_{out,net} = \sum_{out} \dot{m} \left(h + \frac{V^2}{2000} + \frac{gz}{1000} \right) - \sum_{in} \dot{m} \left(h + \frac{V^2}{2000} + \frac{gz}{1000} \right) \quad (6)$$

In Eq. 6, $\dot{Q}$ and $\dot{W}$ are the energy transfer rates by heat and work, h is the enthalpy, V is the velocity, g is acceleration due to gravity, and z is height above datum. Both the compressor and turbine processes are assumed to be adiabatic. For the compressor, kinetic and potential energy differences between the inlet and exit are assumed negligible. However, the turbine exit kinetic energy needs to be included since the exhaust duct is not a significant nozzle and the exhaust gas contributes to engine power output as useful kinetic energy that generates thrust. The publication turbine energy balance did not include this kinetic energy, even though they included it in the total power output for the engine. Equations for shaft power input for the compressor and shaft power output for the turbine are

$$\dot{W}_{AC} = \dot{m}_1(h_2 - h_1) \quad (7)$$

$$\dot{W}_{GT} = \dot{W}_7 = \dot{m}_4(h_4 - h_5) + \dot{m}_{2,2}(h_{2,2} - h_{2,3}) - (\dot{m}_4 + \dot{m}_{2,2}) \frac{V_5^2}{2000} \quad (8)$$

Calculation of the difference in enthalpies, as in Eq. 7 and 8, is discussed in Appendix III (page 15).

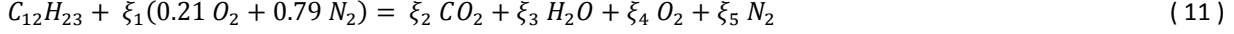
An energy balance was performed on the combustion chamber to assess combustion efficiency. Given that this is a combustion reaction with different reactants than products, the approach taken was to first mix the reactants to obtain a reactants mixture temperature, then bring the products temperature down to the reactants mixture temperature with the removal of heat. The rate of this specific heat transfer is then compared to the lower heating value (LHV) of the fuel to obtain an indication of combustion efficiency.

$$T_{mix} = \frac{\dot{m}_{2,1}C_{p,a}T_{2,1} + \dot{m}_3C_{p,F}T_3}{\dot{m}_{2,1}C_{p,a} + \dot{m}_3C_{p,F}} \quad (9)$$

$$\eta_{l,cc} = \frac{\int_{T_{mix}}^{T_4} c_{p,cg}(T)dT}{LHV} \quad (10)$$

where C_F is the specific heat of the fuel, taken as 2.0 kJ/kg·K [3], and T is temperature. The fuel is JP-8 jet fuel, like kerosene with a chemical formula of $C_{12}H_{23}$, and with a LHV of 42,800 kJ/kg [1]. Although not used directly in the exergy analysis, the combustion efficiency was calculated for general interest and found to be 88.5% for 75%-mode.

A simplified combustion equation was balanced for the four scenarios of air to fuel ratio (AF) to analyze mole fractions of products for obtaining $c_{p,cg}(T)$. It assumes the reaction is complete:



where ξ is the number of moles for each substance.

For the exhaust duct, the energy balance needs to include heat transfer (rationale discussed in section 3.3) and the kinetic energy components. The equation is solved for the velocity of the turbine exit, or exhaust duct inlet:

$$V_5 = \sqrt{\frac{(\dot{m}_{2.3} + \dot{m}_5) V_6^2 + 2000 [\dot{m}_{2.3}(h_{2.4} - h_{2.3}) + \dot{m}_5(h_6 - h_5) + \dot{Q}_{ED,out}]}{(\dot{m}_{2.3} + \dot{m}_5)}} \quad (12)$$

In Eq. 12, $\dot{Q}_{ED,out}$ is an unknown and solution of the equation is postponed until a second law analysis is introduced. V_6 ($=V_{2.4}$) is calculated using the measured thrust of the engine and the relationship:

$$F = \dot{m}_{2.4} V_{2.4} + \dot{m}_6 V_6 - \dot{m}_1 V_1 \quad (13)$$

with F being the thrust by the engine on the test stand. It is assumed the inlet velocity (V_1) is zero.

For the GTMS and gear box, the energy balance includes only shaft power terms and associated power loss due to inefficiencies caused mainly by friction. This friction manifests as heat transfer in the energy balance.

$$\dot{W}_{11} = \eta_{I,GTMS} \dot{W}_7 - \dot{W}_{AC} - \dot{W}_9 \quad (14)$$

$$\dot{W}_{12} = \eta_{I,GB} \dot{W}_{11} \quad (15)$$

The efficiencies of the GTMS and gear box components, although unique, in this analysis are assumed to be equal since shaft power at the gear box inlet was not physically measured. This assumption allows determination of $\dot{W}_{11}$ and $\eta_{I,GTMS}(=\eta_{I,GB})$, which could otherwise not be found. Furthermore, these efficiencies are equivalent to the second law efficiencies for those components since energy transfer by work is equivalent to exergy transfer by work. Hence, $\eta_{II,GTMS} = \eta_{II,GB} = \eta_{I,GTMS} = \eta_{I,GB}$.

3.3 Second Law Analysis

The exergy balance for a component control volume, assuming steady-flow, is

$$\sum \left(1 - \frac{T_0}{T_k}\right) \dot{Q}_{k,in} + \dot{W}_{in} + \sum_{in} \dot{X}_f - \dot{X}_D = \sum \left(1 - \frac{T_0}{T_k}\right) \dot{Q}_{k,out} + \dot{W}_{out} + \sum_{out} \dot{X}_f \quad (16)$$

where T_0 is the surrounding environment temperature, T_k is the temperature at the boundary where heat transfer $\dot{Q}_k$ takes place, $\dot{X}_f$ is the flow exergy rate (exergy transfer by mass entering and leaving the control volume), and $\dot{X}_D$ is the rate of exergy destruction within the control volume. Also, $\dot{X}_f$ at any state throughout the TPE is given by

$$\dot{X}_f = \dot{m} \left[(h - h_0) - T_0(s - s_0) + \frac{V^2}{2000} + \frac{gz}{1000} \right] \quad (17)$$

where h_0 and s_0 are the specific enthalpy and specific entropy of the surrounding environment, and h and s are the specific enthalpy and specific entropy of the state being investigated. The general formula for the second law efficiency, η_{II} , is

$$\eta_{II} = \frac{\dot{X}_{recovered}}{\dot{X}_{expended}} = \frac{\dot{X}_{expended} - \dot{X}_D}{\dot{X}_{expended}} = 1 - \frac{\dot{X}_D}{\dot{X}_{expended}} \quad (18)$$

For the compressor, kinetic and potential exergy changes across the control volume are assumed negligible. For the turbine, potential energy changes across the control volume are assumed negligible but the exit velocity is included as discussed earlier. Second law efficiency and destroyed exergy for these two components are given by

$$\eta_{II,AC} = \frac{\dot{X}_{f,2} - \dot{X}_{f,1}}{\dot{W}_{AC}} \quad (19)$$

$$\dot{X}_{D,AC} = \dot{W}_{AC} + \dot{X}_{f,1} - \dot{X}_{f,2} \quad (20)$$

$$\eta_{II,GT} = \frac{\dot{W}_7}{(\dot{X}_{f,4} - \dot{X}_{f,5}) + (\dot{X}_{f,2,2} - \dot{X}_{f,2,3})} \quad (21)$$

$$\dot{X}_{D,GT} = (\dot{X}_{f,4} - \dot{X}_{f,5}) + (\dot{X}_{f,2,2} - \dot{X}_{f,2,3}) - \dot{W}_7 \quad (22)$$

The combustion chamber exergetic efficiency was analyzed by modeling the expended exergy, or exergy input (fuel), as an exergy transfer by mass and heat.

$$\eta_{II,CC} = \frac{\dot{X}_{f,4} - \dot{X}_{f,2,1} - \dot{X}_{f,3}}{\left(1 - \frac{T_0}{T_{Ad}}\right) \dot{Q}_{chem,F}} \quad (23)$$

$$\dot{Q}_{chem,CC} = \dot{m}_3 LHV \quad (24)$$

$$\dot{X}_{D,CC} = \left(1 - \frac{T_0}{T_{Ad}}\right) \dot{Q}_{chem,F} + \dot{X}_{f,3} + \dot{X}_{f,2,1} - \dot{X}_{f,4} \quad (25)$$

where T_{Ad} is the adiabatic flame temperature, and $\dot{Q}_{chem,F}$ is the equivalent heat transfer rate of the fuel. The calculated $\dot{Q}_{chem,F}$ is the maximum amount of heat transferred for the case of stoichiometric and complete combustion. T_{Ad} was calculated using the NASA Chemical Equilibrium with Applications (CEA) website with the stoichiometric air to fuel ratio (AF) of 14.6 and verified by hand calculation. The output file with the CEA results is included in Appendix IV (page 16). Actual T_{Ad} is lower than the stoichiometric T_{Ad} due to the actual AF being between 40 and 60 (excess air). Also, actual $\dot{Q}_{chem,F}$ is lower than that given by Eq. 24 due to incomplete combustion. Note that there are ten chemical reaction products estimated by CEA as opposed to the four assumed in the more simplified complete reaction used to estimate mole fractions for the calculation of average C_p for combustion gas.

The above method for estimating the combustion chamber expended exergy gives full credit to the potential of the fuel for use in any application. This method contrasts with methods used by others [1, 4, 5] wherein the expended exergy is obtained by using only chemical properties of the fuel. The comparison of the two is saved for Section 4.

An entropy balance for the exhaust duct is given by

$$\dot{\sigma}_{ED} = \dot{m}_5(\dot{s}_6 - \dot{s}_5) + \dot{m}_{2,3}(\dot{s}_{2,4} - \dot{s}_{2,3}) + \frac{\dot{Q}_{ED,out}}{T_5} \quad (26)$$

In this equation $\dot{\sigma}_{ED}$ is the entropy generation for the exhaust duct. The duct was initially assumed to be adiabatic; however, Eq. 26 gave a negative result for entropy generation, which is impossible. One reason could be that $T_{2,4}$ and $T_{2,4}$

(duct exit temperatures) are incorrect. A more plausible explanation is that the duct, being made of metal and exposed to the atmosphere without insulation, is transferring heat to the environment. Assuming the latter, and given the three unknowns of $V_5(=V_{2.3})$, $\dot{Q}_{ED,out}$, and $\dot{\sigma}_{ED}$ with only two equations (energy and exergy balance), it was assumed that $\dot{\sigma}_{ED} = 0$. This is not an unreasonable assumption given the minor change in temperature and pressure across the duct. With this, Equations 12 and 24 can be solved for $V_5(=V_{2.3})$ and $\dot{Q}_{ED,out}$ respectively. The efficiency of the exhaust duct was analyzed assuming it is a nozzle, albeit an ineffective one. Exergy efficiency and exergy destruction rate for the duct are:

$$\eta_{II,ED} = \frac{\dot{m}_{2.3}(h_{2.3} - h_{2.4}) + \dot{m}_5(h_5 - h_6) - \dot{Q}_{ED,out}}{\dot{m}_{2.3}[(h_{2.3} - h_{2.4}) - T_0(s_{2.3} - s_{2.4})] + \dot{m}_5[(h_5 - h_6) - T_0(s_5 - s_6)] - \left(1 - \frac{T_0}{T_5}\right) \dot{Q}_{ED,out}} \quad (27)$$

$$\dot{X}_{D,ED} = (\dot{X}_{f,2.3} - \dot{X}_{f,2.4}) + (\dot{X}_{f,5} - \dot{X}_{f,6}) - \left(1 - \frac{T_0}{T_5}\right) \dot{Q}_{ED,out} \quad (28)$$

Second law efficiency of the GTMS and gear box is reduced to a first law efficiency as discussed in Section 3.2. Together with exergy destruction rate, they are

$$\eta_{II,GTMS} = \frac{\dot{W}_9 + \dot{W}_{11} + \dot{W}_{AC}}{\dot{W}_7} \quad (29)$$

$$\eta_{II,GB} = \frac{\dot{W}_{12}}{\dot{W}_{11}} \quad (30)$$

$$\dot{X}_{D,GTMS} = \dot{W}_7 - (\dot{W}_9 + \dot{W}_{11} + \dot{W}_{AC}) \quad (31)$$

$$\dot{X}_{D,GB} = \dot{W}_{11} - \dot{W}_{12} \quad (32)$$

For the turboprop engine as a whole, the exergetic efficiency is the ratio of combined propeller shaft power ($\dot{W}_{12}$) and kinetic exergy of the exhaust gases to the chemical exergy rate of the fuel:

$$\eta_{II,TPE} = \frac{\dot{W}_{PS,eq}}{\dot{X}_{chem,F}} = \frac{\dot{W}_{12} + \dot{X}_{ke,EG}}{\dot{X}_{chem,F}} = \frac{\dot{W}_{12} + \frac{F^2}{2000(\dot{m}_{2.4} + \dot{m}_6)}}{\left(1 - \frac{T_0}{T_{Ad}}\right) \dot{Q}_{chem,F}} \quad (33)$$

where $\dot{W}_{PS,eq}$ is the equivalent propeller shaft power and $\dot{X}_{ke,EG}$ is the kinetic exergy of the exhaust gases. Kinetic exergy of the exhaust gases is included as part of the recovered exergy because it contributes to thrust, like the propeller. Generally, for an aircraft engine the propulsive efficiency would be used for the exergetic efficiency because the propulsive power of the engine is ultimately the useful engine power. The propulsive power is given by $\dot{W}_p = F V = \dot{m} (V_{exit} - V_{inlet}) V_{aircraft}$. However, for a static engine test the propulsive power is zero. Exergy destruction rate for the engine is

$$\dot{X}_{D,TPE} = \dot{X}_{chem,F} - \dot{W}_{12} - \dot{X}_{ke,EG} \quad (34)$$

4. Results and Comparison to Publication

Numerical results of this analysis are included in Appendix V (page 18). MATLAB was used for all computation, and scripts are included in Appendix VI (page 20). MS Excel was used mainly for summary tables. Figures 3 through 5 provide visual results for the exergetic efficiencies for the four turboprop engine operation modes in this study. Figure 3 compares this study to the publication for the whole turboprop engine and is a direct comparison to Table 1:

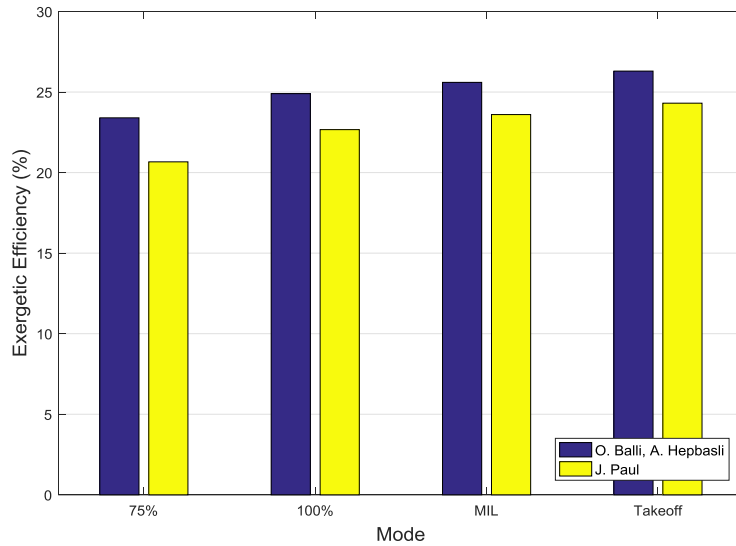


Fig 3. Comparison of turboprop engine exergetic efficiency for four operation modes.

One observation is that the efficiency increases with application of more power—the equivalent shaft power output increases relatively more than the increase in fuel exergy rate. A likely reason for this is that since the increase in fuel exergy rate increases the turbine inlet temperature, increased power output of the turbine more than offsets the higher fuel mass flow rate. (Recall that adiabatic flame temperature is a constant for all operation modes.) A second observation is the difference in exergetic efficiency between this study and the publication of 2-3%. One reason is that the fuel exergy rate is calculated differently in both studies as discussed previously. The other is that, due to mathematically inaccurate formulation of enthalpy change in the publication, there are material differences in the calculations for both compressor shaft power input and turbine shaft power output between the two studies. In this study, it was found that the calculated shaft power input to the gear box was lower than the measured propeller shaft power. Obviously, if the measured propeller shaft power is accurate, this suggests one or more of the other measurements are incorrect. Since the source of the data could not be verified, the propeller shaft power was corrected to match the calculated result, assuming mechanical efficiency of the GTMS and gear box are 100%. Possible suspects for incorrect measurement are the air and fuel flow rates. Increasing fuel flow rate increases the equivalent propeller shaft power output but decreases the efficiency. Increasing air flow rate increases both the equivalent propeller shaft power and the efficiency. Proportionally increasing both fuel and air flow rates increases both the equivalent propeller shaft power and maintains the efficiency relatively constant as expected; however, to match the publication equivalent propeller shaft power requires an increase to the fuel and air flow rates of about 30%. Fig. 4 shows the individual component results for this study:

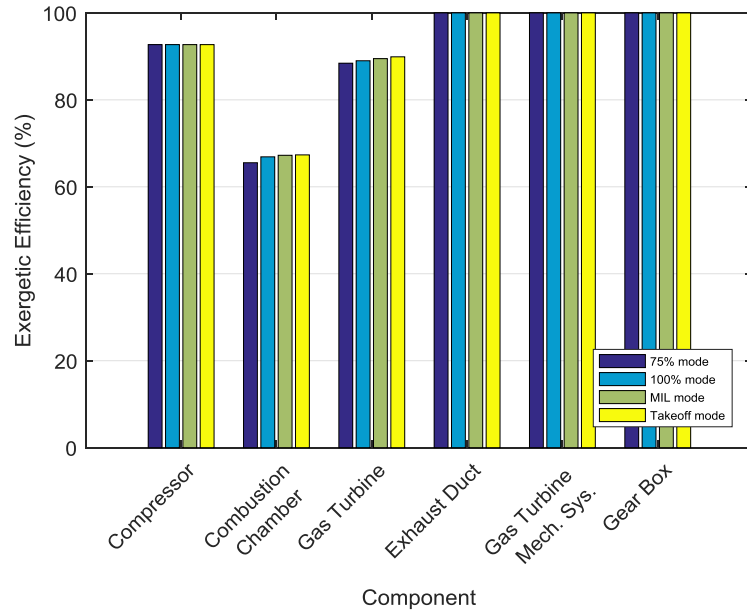


Fig 4. Exergetic efficiency of the turboprop engine components for four operation modes.

The differences in exergetic efficiency between the operating modes for each component are subtle. For the compressor, the efficiency values are constant. This makes sense since the turboprop engine speed was maintained constant throughout the test, and the compressor does not “see” the effects of increasing fuel flow rate. In fact, the temperature and pressure measurement data at the inlet and exit of the compressor are the same for each operating mode. For both the combustion chamber and the turbine, the efficiency increases slightly. For the turbine, the increase in efficiency is again likely due to the increase in turbine inlet temperature. Fig. 5 compares this study to the publication for the individual components and is a direct comparison to Fig. 1:

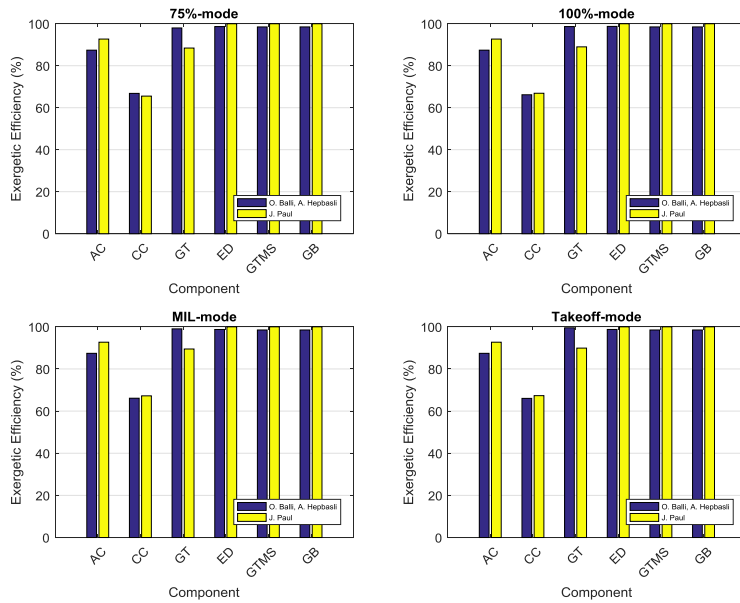


Fig 5. Comparison of component exergetic efficiency for four operation modes.

In general, there are minor differences between the two studies in the component efficiency results, with the largest being for the turbine. Both the compressor and the turbine incur differences due to the way in which the publication calculates enthalpy change as mentioned previously. For the combustion chamber, the efficiency is calculated differently:

This Study (Eq. 23 with $\dot{X}_{f,3}=0$)	Publication
$\eta_{II,CC} = \frac{\dot{X}_{f,4} - \dot{X}_{f,2.1}}{\dot{X}_{chem,F}}$	$\eta_{II,CC} = \frac{\dot{X}_{f,4}}{\dot{X}_{chem,F} + \dot{X}_{f,2.1}}$

Aside from the difference in the way fuel exergy rate is calculated as previously discussed, the publication adds the inlet air exergy rate to the expended exergy term. However, the net effect of both methodology differences is a relatively minor difference in efficiency. In this study, the exergetic efficiency of the duct is 100% since the entropy generation was set to zero as discussed earlier. In the publication, exergetic efficiency was calculated as the ratio of exergy leaving the duct to exergy entering.

5. Turboprop Engine Improvements

An indication as to how the fuel exergy is consumed throughout the engine (or dispelled as exhaust gas) is illustrated in the following figure for the 75%-mode of operation:

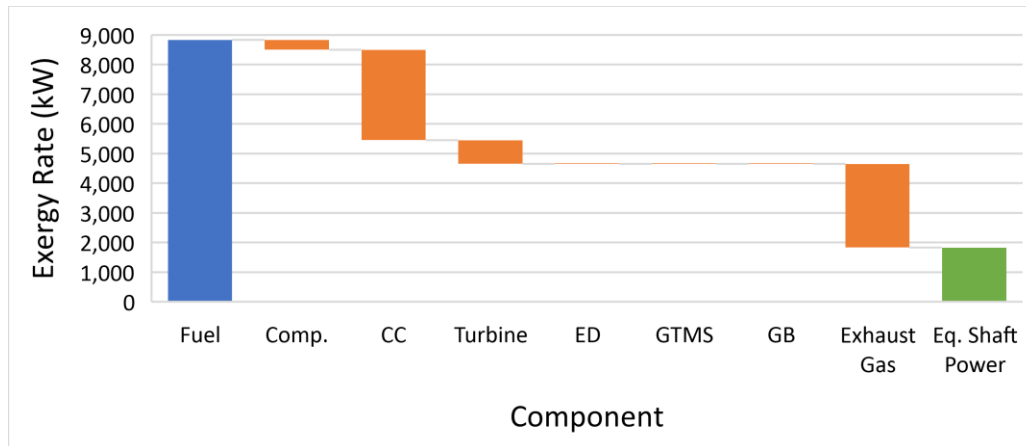


Fig 6. Fuel exergy rate, exergy destruction rate, and power output balance for 75%-mode.

Similarly, the next figure gives an indication to the relative amounts of exergy destruction.

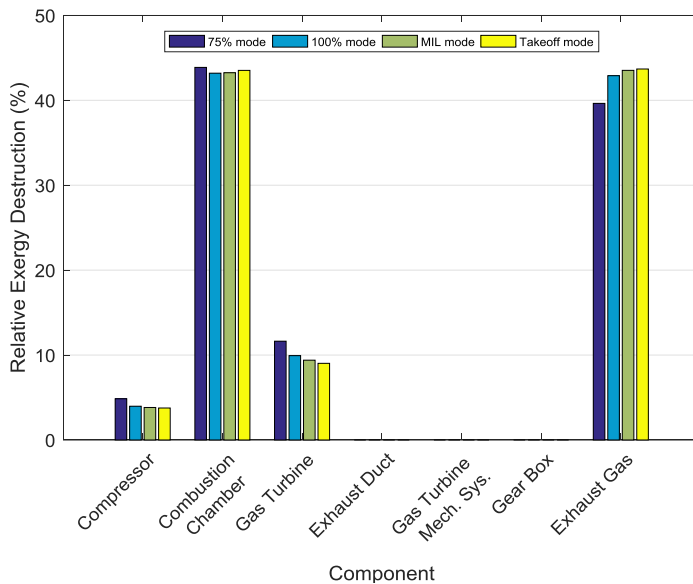


Fig 7. Relative exergy destruction for the engine components and exhaust gas.

The combustion chamber and exhaust gas are responsible for most of the exergy destruction. It's not feasible to recover any of the exhaust gas exergy other than the kinetic exergy component used to generate thrust. The leftover exergy in these gases dissipate into the environment. Instead, this study shows that the combustion chamber should be a primary focus for design improvement efforts to reduce exergy destruction. In fact, the T56 turboprop engine is a 1960's design and improvements to combustion chamber efficiency have been made and incorporated into more recent engine designs [1]. Other turbofan engine analyses yield similar conclusions regarding low combustion chamber exergetic efficiency [4, 5], with the range in values of 58-76%. Higher combustion chamber efficiencies are obtained with better materials and coatings, allowing for higher flame temperatures, for example. However, combustion in general is a highly irreversible process. Higher flame temperatures would require that turbine blades be made more resistant to the higher temperatures with a combination of using super-alloy materials and blade cooling technologies.

6. Exergy Analysis Improvements and Lessons Learned

Several simplifications were made throughout the course of the study. One was in the assumption that the working fluids are ideal gases. Departure from ideal gas behavior becomes significant for high density values [3]. The degree of departure could be tested for the turbine inlet conditions.

The method used in this study for estimating the chemical exergy of the fuel is likely a simplification to generally accepted methods. For example, this property was calculated using a different technique in several publications [1, 4, 5].

In this study, calculations for exergy rates at each of the states suggested that, following a balance of exergy rates, some of the measurement data are inaccurate. It would be beneficial to be able to determine the source of the errors. A sensitivity was run by proportionally increasing both the mass and air flow rates to obtain a match to the measured propeller shaft power output, suggesting that if the measured flow rates are in error, they could be too low by as much as 30%. The publication assessed compressor and turbine shaft power using first law energy balance, but calculating enthalpy change in a manner that introduces material error (7 to 13%) compared to using ideal gas tables. Coincidentally, this introduced error yields a more favorable exergy balance when comparing results to the measured propeller shaft power.

The cooling air for the turbine blades was treated as a secondary flow stream entering and exiting the turbine, with the full mass flow rate entering the turbine in parallel with the combustion gas (similar to the publication). In reality, if an internal blade cooling approach is used, the cooling air is introduced gradually through the turbine stages since it passes through the inside of all of the blades, eventually mixing with the combustion gas. The cooling air serves to decrease the turbine power output, and I estimate that detrimental impact of the cooling air on turbine performance is likely half of what this study reflects; therefore, more work is necessary to handle the cooling air more accurately.

7. Conclusions

An exergy analysis was performed on the T56 turboprop engine and components for four operating modes. Exergetic efficiencies of the entire engine for the four operating modes ranged from 20.7 to 24.3% and were about 3% lower than the publication results. Component exergetic efficiencies were also compared to the publication with differences being the most noticeable for the turbine and compressor. Reasons for the differences were given. The analysis also concluded that a majority of the exergy destruction (other than from the exhaust gas) occurs from the combustion process, and that this should be a focus for engine design improvement. Given this study was conducted within the scope of a course project, several simplifications were made to the exergy analysis as indicated herein. Suggestions for improvements to the analysis process and lessons learned were also given.

8. References

- [1] O. Balli and A. Hepbasli, "Energetic and Exergetic Analysis of T56 Turboprop Engine," *Energy Conversion and Management*, no. 73, pp. 106-120, 2013.
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- [4] Y. Sohret, A. Dinc and H. Karakoc, "Exergy Analysis of a Turbofan Engine for an Unmanned Aerial Vehicle During a Surveillance Mission," *Energy*, no. 93, pp. 716-729, 2015.
- [5] O. Turan, "An exergy way to quantify sustainability metrics for a high bypass turbofan engine," *Energy*, no. 86, pp. 722-736, 2015.

Appendix I

Nomenclature

Latin symbols

C_p	specific heat at constant pressure (kJ/kg·K)
F	thrust (kN)
g	acceleration due to gravity (m/s ²)
h	specific enthalpy (kJ/kg)
LHV	lower heating value of fuel (kJ/kg)
$\dot{m}$	mass flow rate (kg/s)
P	pressure (kPa)
$\dot{Q}$	heat transfer rate (kW)
s	specific entropy (kJ/kg·K)
T	temperature (K)
V	velocity (m/s)
$\dot{W}$	power (kW)
z	elevation above datum (m)

Greek symbols

η_I	energetic efficiency (%)
η_{II}	exergetic efficiency (%)
ξ	combustion equation constant
$\dot{X}$	exergy rate (kW)

Abbreviations and Subscripts

O	environment state
a	air
AC	air compressor
Acc	accessories
Ad	adiabatic
CC	combustion chamber
cg	combustion gas
chem	chemical
D	destruction
ED	exhaust duct
EG	exhaust gas
eq	equivalent
f	flow (as in flow exergy)
F	fuel
GB	gearbox
GT	gas turbine
GTMS	gas turbine mechanical shaft
in	inlet or input
k	the k^{th} component
ke	kinetic energy or kinetic exergy
mix	mixture
out	outlet or output
PS	propeller shaft
TPE	turboprop engine

Appendix II

Measurement Data

The following measurement data are obtained from the turboprop engine test [1]. Balli and Hepbasli [1] combined measurement data with analysis data in their Tables 4 through 7, whereas data included below are measurement data only. Also, in their Tables 6 and 7, mass flow rate for states 5 and 6 are incorrect (looks like a formula-copy-down error).

Measurement data at 75%-mode.

State No.	Fluid type/work	Mass Flow Rate (kg/s)	Temperature (K)	Pressure (kPa)	Shaft Power (kW)	Force (lbf)
0	Air	-	298.15	93.60	-	-
1	Air	14.750	-	-	-	-
2	Air	-	602.25	884.52	-	-
3	Fuel	0.235	298.15	220.64	-	-
4	Combustion gas	-	1,165.15	857.98	-	-
5, 2.3	Combustion gas, cooling air	-	741.00	95.35	-	-
6, 2.4	Combustion gas, cooling air	-	737.30	94.87	-	-
12	Shaft power	-	-	-	2198.7	-
TPE	Residual thrust	-	-	-	-	688

Measurement data at 100%-mode.

State No.	Fluid type/work	Mass Flow Rate (kg/s)	Temperature (K)	Pressure (kPa)	Shaft Power (kW)	Force (lbf)
0	Air	-	298.15	93.60	-	-
1	Air	14.750	-	-	-	-
2	Air	-	602.25	884.52	-	-
3	Fuel	0.294	298.15	220.64	-	-
4	Combustion gas	-	1,285.15	857.98	-	-
5, 2.3	Combustion gas, cooling air	-	823.40	95.35	-	-
6, 2.4	Combustion gas, cooling air	-	819.28	94.87	-	-
12	Shaft power	-	-	-	2996.28	-
TPE	Residual thrust	-	-	-	-	735.6

Measurement data at MIL-mode.

State No.	Fluid type/work	Mass Flow Rate (kg/s)	Temperature (K)	Pressure (kPa)	Shaft Power (kW)	Force (lbf)
0	Air	-	298.15	93.60	-	-
1	Air	14.750	-	-	-	-
2	Air	-	602.25	884.52	-	-
3	Fuel	0.309	298.15	220.64	-	-
4	Combustion gas	-	1,315.15	857.98	-	-
5, 2.3	Combustion gas, cooling air	-	840.45	95.35	-	-
6, 2.4	Combustion gas, cooling air	-	836.25	94.87	-	-
12	Shaft power	-	-	-	3256.26	-

TPE	Residual thrust	-	-	-	-	746.1
-----	-----------------	---	---	---	---	-------

Measurement data at Takeoff-mode.

State No.	Fluid type/work	Mass Flow Rate (kg/s)	Temperature (K)	Pressure (kPa)	Shaft Power (kW)	Force (lbf)
0	Air	-	298.15	93.60	-	-
1	Air	14.750	-	-	-	-
2	Air	-	602.25	884.52	-	-
3	Fuel	0.317	298.15	220.64	-	-
4	Combustion gas	-	1,330.22	857.98	-	-
5, 2.3	Combustion gas, cooling air	-	847.00	95.35	-	-
6, 2.4	Combustion gas, cooling air	-	842.77	94.87	-	-
12	Shaft power	-	-	-	3442.01	-
TPE	Residual thrust	-	-	-	-	750

Appendix III

Specific Heats, Change in Specific Enthalpy, and Change in Specific Entropy

The change in specific enthalpy is estimated using the definition of specific heat at constant pressure, combined with the assumption that all gas mixtures in the analysis are assumed to be ideal:

$$\Delta h = \int_a^b c_p(T) dT$$

In the above equation, c_p is the specific heat at constant pressure and T is temperature. $c_p(T)$ was estimated using a polynomial form, as shown below [3] for air and the assumed combustion gas components:

	Temp. Range K	% Error	
		Max	Avg
$c_{p,air}(T) = [28.11 + 0.1967 \times 10^{-2} T + 0.4802 \times 10^{-5} T^2 - 1.966 \times 10^{-9} T^3]/28.97$	273-1800	0.72	0.33
$c_{p,CO_2}(T) = [22.26 + 5.981 \times 10^{-2} T - 3.501 \times 10^{-5} T^2 + 7.469 \times 10^{-9} T^3]/44.01$	273-1800	0.67	0.22
$c_{p,H_2O}(T) = [32.24 + 0.1923 \times 10^{-2} T + 1.055 \times 10^{-5} T^2 - 3.595 \times 10^{-9} T^3]/18.015$	273-1800	0.53	0.24
$c_{p,O_2}(T) = [25.48 + 1.520 \times 10^{-2} T - 0.7155 \times 10^{-5} T^2 + 1.312 \times 10^{-9} T^3]/31.999$	273-1800	1.19	0.28
$c_{p,N_2}(T) = [28.90 - 0.1571 \times 10^{-2} T + 0.8081 \times 10^{-5} T^2 - 2.873 \times 10^{-9} T^3]/28.013$	273-1800	0.59	0.34

Combustion gas mixture properties were calculated using a mass or mole fraction weighted average of components as appropriate.

The change in specific entropy is estimated using the second Gibbs equation and the resulting integral form for obtaining Δs for an ideal gas:

$$T ds = dh - v dP$$

$$s_b - s_a = \int_a^b c_p(T) \frac{dT}{T} - R \ln \frac{P_b}{P_a}$$

In the last equation, the integral is evaluated by maintaining the temperature dependent $c_p(T)$ within the integrand. The above integrals provide a more accurate estimate of Δh and Δs than assuming constant c_p , especially for large temperature differences, and are relatively easy computations using Matlab since $c_p(T)$ is in polynomial form.

Appendix IV

NASA CEA Adiabatic Flame Temperature Results

NASA-GLENN CHEMICAL EQUILIBRIUM PROGRAM CEA2, FEBRUARY 5, 2004
 BY BONNIE MCBRIDE AND SANFORD GORDON
 REFS: NASA RP-1311, PART I, 1994 AND NASA RP-1311, PART II, 1996

```

prob case=paul8299  hp p(bar)=8.5798
o/f=14.63
reac
  fuel Jet-A(L) wt%=100. t,k= 298.15
  oxid  O2          wt%= 23.3 t,k= 602.25
  oxid  N2          wt%= 76.7 t,k= 602.25
  output short
output trace= 1e-5
end
  
```

THERMODYNAMIC EQUILIBRIUM COMBUSTION PROPERTIES AT ASSIGNED

PRESSURES

CASE = paul8299

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	Jet-A(L)	1.0000000	-303403.000	298.150
OXIDANT	O2	0.2330000	9317.276	602.250
OXIDANT	N2	0.7670000	8961.774	602.250

O/F= 14.63000 %FUEL= 6.397953 R,EQ.RATIO= 0.995881 PHI,EQ.RATIO= 0.995881

THERMODYNAMIC PROPERTIES

P, BAR 8.5798
 T, K 2458.81
 RHO, KG/CU M 1.1995 0
 H, KJ/KG 177.15
 U, KJ/KG -538.15
 G, KJ/KG -22052.1
 S, KJ/ (KG) (K) 9.0407

M, (1/n) 28.581
 (dLV/dLP)t -1.00324
 (dLV/dLT)p 1.0894
 Cp, KJ/ (KG) (K) 2.1955
 GAMMAS 1.1821
 SON VEL,M/SEC 919.5

MOLE FRACTIONS

*CO 1.4555-2
 *CO2 1.1660-1
 *H 3.5673-4

*H2	2.4879-3
H2O	1.2113-1
*NO	3.6173-3
*N2	7.3066-1
*O	3.2408-4
*OH	3.7837-3
*O2	6.4926-3

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

Appendix V

Numerical Results

Exergy rate ($\dot{X}_f$ or $\dot{X}_{work}$) and mass flow rate at all states.

State	75%-mode		100%-mode		MIL-mode		Takeoff-mode	
	$\dot{m}$ kg/s	$\dot{X}$ kW	$\dot{m}$ kg/s	$\dot{X}$ kW	$\dot{m}$ kg/s	$\dot{X}$ kW	$\dot{m}$ kg/s	$\dot{X}$ kW
0	0.000	0.0	0.000	0.0	0.000	0.0	0.000	0.0
1	14.750	0.0	14.750	0.0	14.750	0.0	14.750	0.0
2	14.750	4,278.1	14.750	4,278.1	14.750	4,278.1	14.750	4,278.1
2.1	13.275	3,850.3	13.275	3,850.3	13.275	3,850.3	13.275	3,850.3
2.2	1.475	427.8	1.475	427.8	1.475	427.8	1.475	427.8
2.3	1.475	297.3	1.475	383.4	1.475	402.1	1.475	409.3
2.4	1.475	294.7	1.475	380.3	1.475	398.8	1.475	406.0
3	0.235	0.0	0.294	0.0	0.309	0.0	0.317	0.0
4	13.510	9,642.6	13.569	11,246.7	13.584	11,665.1	13.592	11,878.7
5	13.510	2,794.5	13.569	3,642.2	13.584	3,829.9	13.592	3,904.1
6	13.510	2,769.7	13.569	3,611.6	13.584	3,798.1	13.592	3,871.9
7	-	6,170.8	-	6,807.1	-	7,033.8	-	7,185.5
8	-	6,170.8	-	6,807.1	-	7,033.8	-	7,185.5
9	-	41.8	-	41.8	-	41.8	-	41.8
10	-	6,129.0	-	6,765.4	-	6,992.0	-	7,143.8
11	-	1,514.1	-	2,150.4	-	2,377.1	-	2,528.9
12	-	1,514.1	-	2,150.4	-	2,377.1	-	2,528.9

Exergetic efficiency and other thermodynamic parameters of the engine components.

Component	75%-mode				100%-mode			
	η_{II} %	$\dot{Q}_{out}$ kW	$\dot{W}_{out}$ kW	$\dot{X}_D$ kW	η_{II} %	$\dot{Q}_{out}$ kW	$\dot{W}_{out}$ kW	$\dot{X}_D$ kW
Compressor	92.7%	-	-4,614.9	336.9	92.7%	-	-4,614.9	336.9
Combustion Chamber	65.5%	-	-	3,046.1	66.9%	-	-	3,661.0
Turbine	88.4%	-	6,170.8	807.7	89.0%	-	6,807.1	841.8
Exhaust Duct	100.0%	45.9	-	0.0	100.0%	52.9	-	0.0
Gas Turb. Mech. Sys.	100.0%	-	-	0.0	100.0%	-	-	0.0
Gear Box	100.0%	-	-	0.0	100.0%	-	-	0.0
Exhaust Gas	-	-	-	2,821.1	-	-	-	3,711.5
Component	MIL-mode				Takeoff-mode			
	η_{II} %	$\dot{Q}_{out}$ kW	$\dot{W}_{out}$ kW	$\dot{X}_D$ kW	η_{II} %	$\dot{Q}_{out}$ kW	$\dot{W}_{out}$ kW	$\dot{X}_D$ kW
Compressor	92.7%	-	-4,614.9	336.9	92.7%	-	-4,614.9	336.9
Combustion Chamber	67.2%	-	-	3,806.7	67.3%	-	-	3,894.0
Turbine	89.5%	-	7,033.8	827.2	89.9%	-	7,185.5	807.5
Exhaust Duct	100.0%	54.3	-	0.0	100.0%	54.8	-	0.0
Gas Turb. Mech. Sys.	100.0%	-	-	0.0	100.0%	-	-	0.0
Gear Box	100.0%	-	-	0.0	100.0%	-	-	0.0
Exhaust Gas	-	-	-	3,908.0	-	-	-	3,985.8

Exergetic efficiency and other performance parameters of the turboprop engine.

Parameter	Units	Operation Mode			
		75%	100%	MIL	Takeoff
T_{Ad}	K	2,458.8	2,458.8	2,458.8	2,458.8
$\dot{Q}_{chem,CC}$	kW	8,838.4	11,057.4	11,621.5	11,922.4
$\dot{W}_{12}$	kW	1,514.1	2,150.4	2,377.1	2,528.9
$\dot{X}_{ke,EG}$	kW	312.5	355.8	365.7	369.4
$\dot{W}_{PS,eq}$	kW	1,826.6	2,506.3	2,742.8	2,898.2
$\dot{X}_{D,TPE}$	kW	7,011.7	8,551.1	8,878.7	9,024.2
$\eta_{II,TPE}$		20.7%	22.7%	23.6%	24.3%

Parameters of the combustion reaction, including gas constant for combustion gas.

Constant	Units	Operation Mode			
		75%	100%	MIL	Takeoff
Air:Fuel		56.5	45.2	43.0	41.9
ξ_1	kmol	326.2	260.8	248.1	241.9
ξ_2	kmol	12.0	12.0	12.0	12.0
ξ_3	kmol	11.5	11.5	11.5	11.5
ξ_4	kmol	50.8	37.0	34.4	33.0
ξ_5	kmol	257.7	206.0	196.0	191.1
R	kJ/kg·K	0.288137	0.288126	0.288124	0.288122

Appendix VI

MATLAB Scripts

```

% Main script for exergy efficiency calculations and plotting
%
clc
clear variables
close all
format compact

Rair = 0.287; % gas constant for air, kJ/Kg·K
CpKero = 2.0; % Cp for kerosene @ 20°C, kJ/Kg·K; Cengel/Boles A-3
LHVkero = 42800; % lower heating value for kerosene, KJ/kg; focus paper

% raw data from the paper is stored in structure array and saved to file
% struct has 4 sets of fields—one set for each operation mode
% operation modes: 75%-mode, 100%-mode, MIL-mode, Takeoff-mode
load paper_data.mat

% adiabatic flame temperatures from NASA CEA online calculator
% values reflect excess air
Tad=[2458.81; 2458.81; 2458.81; 2458.81]; % one for each mode, K

% allocate memory for calculated results
eta_II=zeros(6,4); % exergetic efficiency
X_dot_d=zeros(8,4); % exergy destruction for 6 components + whole TPE
W_dot_eq=zeros(1,4); % power output equivalent (shaft + KE exhaust gas)
X_dot_chemfuel=zeros(1,4); % fuel chemical exergy rate modeled as heat transfer

for mode=1:4
    % mass flow rates (kg/s): vector of values, one for each state
    m_dot=paper_rawdata(mode).m_dot;
    % overwrite values of m_dot to experiment with matching W_dot_12
    m_dot(2)=m_dot(2)*1.0; % set to manual value, otherwise m_dot(2)
    m_dot(3)=m_dot(2);
    m_dot(4)=0.9*m_dot(3);
    m_dot(5)=0.1*m_dot(3);
    m_dot(6)=m_dot(5);
    m_dot(7)=m_dot(6);
    m_dot(8)=m_dot(8)*1.0; % set to manual value, otherwise m_dot(8)
    m_dot(9)=m_dot(4)+m_dot(8);
    m_dot(10)=m_dot(9);
    m_dot(11)=m_dot(10);

    % temperature (K): vector of values, one for each state
    T=paper_rawdata(mode).T;
    % pressure (kPa): vector of values, one for each state
    P=paper_rawdata(mode).P;

    % calculate stoichiometric # moles of combustion gas components
    y=prod_moles(m_dot(4), m_dot(8));

    Rcg = Rcombgas(y); % gas constant for combustion gas, kJ/Kg·K

    % allocate memory for calculated results
    X_dot=zeros(11,1); % Exergy rate
    W_dot=zeros(6,1); % shaft power

    % power to accessories is assumed constant for all modes
    W_dot(3) = 41.76;
    % add measured propeller shaft power to W_dot vector
    W_dot(6) = paper_rawdata(mode).W12_dot;

    % measured force on TPE (exhaust gas vel. only), converted from lbf to N

```

```

F=paper_rawdata(mode).F * 4.448222;

% XXXXXXXXXXXX ENERGY AND EXERGY ANALYSIS XXXXXXXXXXXX

% state 0
X_dot(1)=0; % at environment
% XXXXXXXXXXXX Compressor XXXXXXXXXXXX
% state 1 exergy rate
X_dot(2)=0; % at environment
% state 2 exergy rate
X_dot(3)=m_dot(3)*psi_air(Rair,T(1),T(3),P(1),P(3),0,0);
% 1st law energy balance to get shaft power input
W_dot_c = m_dot(2)*int_CpAir_dT(T(1),T(3));
% exergetic efficiency
eta_II(1,mode) = (X_dot(3) - X_dot(2)) / W_dot_c;
% exergy destruction
X_dot_d(2,mode)=X_dot(2)+W_dot_c-X_dot(3);

% XXXXXXXXXXXX Combustion Chamber XXXXXXXXXXXX
% state 2.1 exergy rate
X_dot(4)=0.9*X_dot(3); % assumed 90% of air flow is for combustion
% state 2.2 exergy rate
X_dot(5)=0.1*X_dot(3); % assumed 10% of air flow is for blade cooling
% state 3 exergy rate-non chem (=0 since T at env T)
X_dot(8)=0;
% state 3 exergy rate (chemical exergy of fuel), modeled as heat accross boundary at Tad
X_dot_chemfuel(mode)=(1-T(1)/Tad(mode))*LHVkero*m_dot(8);
% state 4 exergy rate
X_dot(9)=m_dot(9)*psi_cg(y,Rcg,T(1),T(9),P(1),P(9),0,0);
% exergetic efficiency
eta_II(2,mode) = (X_dot(9)-X_dot(4)-X_dot(8)) / X_dot_chemfuel(mode);
% exergy destruction
X_dot_d(3,mode)=X_dot(8)+X_dot_chemfuel(mode)+X_dot(4)-X_dot(9);

% XXXXXXXXXXXX Exhaust Duct XXXXXXXXXXXXXXXXXXXXXXXX
% adiabatic assumption yields -ve entropy gen->impossible!
% so can't assume adiabatic-> calculate Qdot based on assuming
% entropy gen = 0 since otherwise entropy gen can't be found b/c
% have 3 eqns and 2 unknowns
% find Qdot
Q_dot_ED=T(10)*(m_dot(6)*(delta_s_air(Rair,T(7),T(6),P(7),P(6)))+...
    m_dot(10)*(delta_s_cg(y,Rcg,T(11),T(10),P(11),P(10))));
% calc. vel. at exhaust duct exit using measured force
v6=F/(m_dot(7)+m_dot(11));
v24=v6; % assume cooling air same velocity as comb. gas
% use 1st law balance on exhaust duct to calc. vel. at ED inlet
v5=sqrt((2000*(m_dot(6)*int_CpAir_dT(T(6),T(7))+...
    m_dot(10)*int_CpCG_dT(T(10),T(11),y)+Q_dot_ED)+...
    (m_dot(6)+m_dot(10))*v6^2)/(m_dot(6)+m_dot(10)));
v23=v5; % assume cooling air same velocity as comb. gas
% state 2.3 exergy rate
X_dot(6)=m_dot(6)*psi_air(Rair,T(1),T(6),P(1),P(6),v23,0);
% state 5 exergy rate
X_dot(10)=m_dot(10)*psi_cg(y,Rcg,T(1),T(10),P(1),P(10),v5,0);
% state 2.4 exergy rate
X_dot(7)=m_dot(7)*psi_air(Rair,T(1),T(7),P(1),P(7),v24,0);
% state 6 exergy rate
X_dot(11)=m_dot(11)*psi_cg(y,Rcg,T(1),T(11),P(1),P(11),v6,0);
% exergy destruction
X_dot_d(5,mode)=X_dot(6)+X_dot(10)-X_dot(7)-X_dot(11)-...
    (1-T(1)/T(10))*Q_dot_ED;
% exergetic efficiency
eta_II(4,mode) = (m_dot(6)*int_CpAir_dT(T(7),T(6))+...
    m_dot(10)*int_CpCG_dT(T(11),T(10),y)-Q_dot_ED) /...
    (m_dot(6)*(psi_air(Rair,T(1),T(6),P(1),P(6),0,0)-...
        psi_air(Rair,T(1),T(7),P(1),P(7),0,0))+...
    m_dot(10)*(psi_cg(y,Rcg,T(1),T(10),P(1),P(10),0,0)-...

```

```

        psi_cg(y,Rcg,T(1),T(11),P(1),P(11),0,0))-...
        (1-T(1)/T(10))*Q_dot_ED);

% XXXXXXXXXXXX Turbine XXXXXXXXXXXX
% 1st law energy balance to get shaft power output
W_dot(1) = m_dot(5)*int_CpAir_dT(T(6),T(5))...
        + m_dot(9)*int_CpCG_dT(T(10),T(9),y)-(m_dot(5)+m_dot(9))*v5^2/2000;
% exergetic efficiency
eta_II(3,mode) = W_dot(1)/((X_dot(5)-X_dot(6))+(X_dot(9)-X_dot(10)));
% exergy destruction
X_dot_d(4,mode)=(X_dot(5)-X_dot(6))+(X_dot(9)-X_dot(10))-W_dot(1);

% XXXXXXXXXXXX GTMS and RGB XXXXXXXXXXXXXXXXXXXXXXXX
% handle case where if GTMS efficiency is assumed to be 100% and
% RGB input power is less than measured prop shaft power, then GTMS
% and RGB efficiencies are maxed at 100% (can't exceed 100%).
if W_dot(1)-W_dot(3)-W_dot_c < W_dot(6)
    X_dot_d(6,mode)=0;
    X_dot_d(7,mode)=0;
    W_dot(2) = W_dot(1)-X_dot_d(6,mode);
    W_dot(4) = W_dot(2)-W_dot(3);
    W_dot(5) = W_dot(4)-W_dot_c;
    W_dot(6) = W_dot(5)-X_dot_d(7,mode);
else
    % back calculate W_dot at state 11 (gear box in). Assume efficiencies of
    % GTMS and RGB are equal (same assumption as in paper, since there is
    % no measurement of internal shaft power. Solution is the positive
    % quadratic root.
    W_dot(5) = max(roots([1 W_dot_c+W_dot(3) -W_dot(1)*W_dot(6)]));
    X_dot_d(6,mode)= W_dot(1)-( W_dot(5)+W_dot_c+W_dot(3));
    X_dot_d(7,mode)= W_dot(5)- W_dot(6);
    W_dot(2) = W_dot(1)-X_dot_d(6,mode);
    W_dot(4) = W_dot(1)-X_dot_d(6,mode)-W_dot(3);
end
% eta_II for GTMS
eta_II(5,mode) = (W_dot_c + W_dot(3) + W_dot(5))/W_dot(1);
% eta_II for RGB
eta_II(6,mode) = W_dot(6)/W_dot(5);

% XXXXXXXXXXXX Exhaust gas exergy destruction XXXXXXXXXXXXXXXXXXXXXXXX
% quantify exergy rate of gases leaving exhaust duct. Does not affect
% component exergetic efficiency, but will help with understanding
% exergy destruction from all sources pertaining to the exergetic
% efficiency of whole TPE. Exclude kinetic energy component, which is
% included in exergy recovered for the whole TPE
E_dot_ke = F^2/(2000*(m_dot(7)+m_dot(11))); % kinetic energy of exhaust gas, KW
X_dot_d(8,mode) = X_dot(7)+X_dot(11)-E_dot_ke;

% XXXXXXXXXXXX Overall TPE XXXXXXXXXXXXXXXXXXXXXXXX
W_dot_eq(1,mode)=W_dot(6)+E_dot_ke;
eta_II_TPE(mode)=W_dot_eq(1,mode)/X_dot_chemfuel(mode);
% exergy destruction
X_dot_d(1,mode)=sum(X_dot_d(:,mode));
end

% XXXXXXXXXXXX Plots XXXXXXXXXXXX

% bar chart: etaII vs components; series: modes
figure()
bar(eta_II*100);
xlabel('Component');
ax = gca;
set(ax,'xticklabel',{'Compressor',...
    'Combustion\nexhaust Chamber','Gas Turbine','Exhaust Duct',...
    'Gas Turbine\nexhaust Mech. Sys.','Gear Box'});
ax.YGrid = 'on';
ax.XTickLabelRotation = 45;

```

```

ylabel('Exergetic Efficiency (%)');
ylim([0 100]);
leg=legend('75% mode', '100% mode','MIL mode','Takeoff mode',...
'Location','southeast','Orientation','vertical');
set(leg,'FontSize',10);
clear etaII

% bar chart: whole TPE etaII vs mode; series: paper, me
for i=1:4
    etaII(i,:)=[paper_results(i).eta_II(1) eta_II_TPE(i)];
end
figure()
b=bar(etaII*100);
% b(1).FaceColor = 'blue';
% b(2).FaceColor = 'cyan';
xlabel('Mode');
ax = gca;
set(ax,'xticklabel',{'75%','100%','MIL','Takeoff'});
ax.YGrid = 'on';
ylabel('Exergetic Efficiency (%)');
% ylim([0 100]);
leg=legend('O. Balli, A. Hepbasli', 'J. Paul',...
'Location','southeast','Orientation','vertical');
set(leg,'FontSize',7);
clear etaII

% 1 bar chart per mode: etaII vs TPE component; series: paper, me
figure()
titles={'75%-mode', '100%-mode','MIL-mode','Takeoff-mode'};
for i=1:4
    for j=1:6
        etaII(j,:)=[paper_results(i).eta_II(j+1) eta_II(j,i)];
    end
    subplot(2,2,i)
    b=bar(etaII*100);
    % b(2).FaceColor = 'red';
    title(titles(i));
    xlabel('Component');
    ax = gca;
    set(ax,'xticklabel',{'AC','CC','GT','ED','GTMS','GB'});
    ax.XTickLabelRotation = 45;
    ax.YGrid = 'on';
    ylabel('Exergetic Efficiency (%)');
    leg=legend('O. Balli, A. Hepbasli', 'J. Paul',...
'Location','southeast','Orientation','vertical');
    set(leg,'FontSize',7);
    ylim([0 100])
end
clear etaII

% bar chart: % exergy destruction vs components; series: modes
figure()
bar(X_dot_d(2:8,:)./repelem(sum(X_dot_d(2:8,:)),7,1)*100);
xlabel('Component');
ax = gca;
set(ax,'xticklabel',{'Compressor',...
'Combustion\n Chamber','Gas Turbine','Exhaust Duct',...
'Gas Turbine\n Mech. Sys.','Gear Box','Exhaust Gas'});
ax.YGrid = 'on';
ax.XTickLabelRotation = 45;
ylabel('Relative Exergy Destruction (%)');
ylim([0 50]);
leg=legend('75% mode', '100% mode','MIL mode','Takeoff mode',...
'Location','north','Orientation','horizontal');
set(leg,'FontSize',8);
clear etaII

```

```

function Cp = CpAir(T)
% Calculates Cp for air
% temperature range: 273 - 1800 K
% source: B.G. Kyle, Chemical and Process Thermodynamics
% input: temperature (K)
% output: Cp air (kJ/kgK)

% polynomial coefficients
a=28.11;
b=0.1967e-2;
c=0.4802e-5;
d=-1.966e-9;

Ma=28.97; % molar mass of air

Cp = (a + b*T + c*T.^2 + d*T.^3) / Ma;

```

```

function Cp = CpCG(T,y)
% Calculates Cp for combustion gas with components CO2, H2O, O2, N2
% temperature range: 273 - 1800 K
% source: B.G. Kyle, Chemical and Process Thermodynamics
% inputs: temperature (K)
%          row vector (y) of # moles of the gas components listed above
% output: Cp air (kJ/kgK)

% polynomial coefficients
a=[+22.26    +32.24    +25.48    +28.90];
b=[+5.981e-2 +0.1923e-2 +1.520e-2 -0.1571e-2];
c=[-3.501e-5 +1.055e-5 -0.7155e-5 +0.8081e-5];
d=[+7.469e-9 -3.595e-9 +1.312e-9 -2.873e-9];
% calculate mixture polynomial coefficients
am=sum(y.*a)/sum(y);
bm=sum(y.*b)/sum(y);
cm=sum(y.*c)/sum(y);
dm=sum(y.*d)/sum(y);
% molar mass of components
M=[44.01 18.015 31.999 28.013];
% calculate molar mass of gas mixture
Mcg = sum(y.*M)/sum(y);
% calculate Cp (kJ/kgK)
Cp = (am + bm*T + cm*T.^2 + dm*T.^3) / Mcg;

```

```

function result = delta_s_air(Rair,T0,T1,P0,P1)
% Calculates change in specific entropy at specified state for air
% inputs: Rair:gas constant for air, kJ/Kg·K
%          T1: temperature of state, K
%          T0: temperature of environment, K
%          P1: pressure of state, kPa
%          P0: pressure of environment, kPa
% output: delta s (kJ/kg·K)
result = int_CpAir_divT_dT(T0,T1)-Rair*log(P1/P0);

```

```

function result = delta_s_cg(y,Rcg,T0,T1,P0,P1)
% Calculates change in specific entropy at specified state for combustion gas
% Combustion gas components are assumed to be: CO2, H2O, O2, N2
% inputs: y: row vector (y) of # moles of the gas components listed above
%          Rcg:gas constant for combustion gas, kJ/Kg·K
%          T1: temperature of state, K
%          T0: temperature of environment, K
%          P1: pressure of state, kPa
%          P0: pressure of environment, kPa
% output: output: delta s (kJ/kg·K)

```



```
result = int_CpCG_divT_dT(T0,T1,y)-Rcg*log(P1/P0);
```

```
function int = int_CpAir_divT_dT(T1,T2)
% Calculates integral of Cp/T·dT for air
% temperature range: 273 - 1800 K
% source: B.G. Kyle, Chemical and Process Thermodynamics
% input: endpoint temperatures T1, T2 (K)
% output: integral of Cp/T dt (kJ/kg·K)

% polynomial coefficients
a=28.11;
b=0.1967e-2;
c=0.4802e-5;
d=-1.966e-9;

Ma=28.97; % molar mass of air

int = (a*log(T2/T1) + b*(T2-T1) + c/2*(T2^2-T1^2) + d/3*(T2^3-T1^3))/Ma;
```

```
function int = int_CpAir_dT(T1,T2)
% Calculates integral of Cp·dT for air
% temperature range: 273 - 1800 K
% source: B.G. Kyle, Chemical and Process Thermodynamics
% input: endpoint temperatures T1, T2 (K)
% output: delta h (kJ/kg)

% polynomial coefficients
a=28.11;
b=0.1967e-2;
c=0.4802e-5;
d=-1.966e-9;

Ma=28.97; % molar mass of air

int = (a*(T2-T1) + b/2*(T2^2-T1^2) + c/3*(T2^3-T1^3) + d/4*(T2^4-T1^4))/Ma;
```

```
function int = int_CpCG_divT_dT(T1,T2,y)
% Calculates integral of Cp/T·dT for combustion gas with components CO2, H2O, O2, N2
% temperature range: 273 - 1800 K
% source: B.G. Kyle, Chemical and Process Thermodynamics
% input: endpoint temperatures T1, T2 (K)
%         row vector (y) of # moles of the gas components listed above
% output: integral of Cp/T dt (kJ/kg·K)

% polynomial coefficients
a=[+22.26    +32.24    +25.48    +28.90];
b=[+5.981e-2 +0.1923e-2 +1.520e-2 -0.1571e-2];
c=[-3.501e-5 +1.055e-5 -0.7155e-5 +0.8081e-5];
d=[+7.469e-9 -3.595e-9 +1.312e-9 -2.873e-9];
% calculate mixture polynomial coefficients
am=sum(y.*a)/sum(y);
bm=sum(y.*b)/sum(y);
cm=sum(y.*c)/sum(y);
dm=sum(y.*d)/sum(y);
% molar mass of components
M=[44.01 18.015 31.999 28.013];
% calculate molar mass of gas mixture
M_cg = sum(y.*M)/sum(y);

int = (am*log(T2/T1) + bm*(T2-T1) + cm/2*(T2^2-T1^2) ...
      + dm/3*(T2^3-T1^3))/M_cg;
```

```
function int = int_CpCG_dT(T1,T2,y)
```

```

% Calculates integral of Cp·dT for combustion gas with components CO2, H2O, O2, N2
% temperature range: 273 - 1800 K
% source: B.G. Kyle, Chemical and Process Thermodynamics
% input: endpoint temperatures T1, T2 (K)
%         row vector (y) of # moles of the gas components listed above
% output: delta h (kJ/kg)

% polynomial coefficients
a=[+22.26      +32.24      +25.48      +28.90];
b=[+5.981e-2 +0.1923e-2 +1.520e-2  -0.1571e-2];
c=[-3.501e-5 +1.055e-5  -0.7155e-5 +0.8081e-5];
d=[+7.469e-9 -3.595e-9  +1.312e-9  -2.873e-9];
% calculate mixture polynomial coefficients
am=sum(y.*a)/sum(y);
bm=sum(y.*b)/sum(y);
cm=sum(y.*c)/sum(y);
dm=sum(y.*d)/sum(y);
% molar mass of components
M=[44.01 18.015 31.999 28.013];
% calculate molar mass of gas mixture
Mcg = sum(y.*M)/sum(y);

int = (am*(T2-T1) + bm/2*(T2^2-T1^2) + cm/3*(T2^3-T1^3)...
      + dm/4*(T2^4-T1^4)) / Mcg;

```

```

function y = prod_moles(m_dot_air, m_dot_fuel)
% generates # kmol of each combustion product per 1 kmol of reactant fuel
% inputs: mass flow rate of air, mass flow rate of fuel
% output: row vector of # moles of each product: CO2, H2O, O2, N2
%
% molar masses (kg/kmol)
Mf=167.311; % C12H23 JP-8 jet fuel, similar to kerosene
Ma=28.97; % molar mass of air

AF=m_dot_air/m_dot_fuel; % air to fuel ratio
xi1=AF*Mf/Ma; % # moles air required per mole fuel for complete combustion
xi2=12; % # moles CO2 produced per mole fuel in complete combustion
xi3=11.5; % # moles H2O produced per mole fuel in complete combustion
xi4=(xi1*0.21*2-xi2*2-xi3*1) / 2; % # moles O2 produced per mole fuel in complete combustion
xi5=xi1*0.79; % # moles N2 produced per mole fuel in complete combustion
% disp( [xi1, xi2, xi3, xi4, xi5]);
y=[xi2 xi3 xi4 xi5];

```

```

function result = psi_air(Rair,T0,T1,P0,P1,v,z)
% Calculates exergy rate at specified state air
% inputs: Rair:gas constant for air, kJ/Kg·K
%         T1: temperature of state, K
%         T0: temperature of environment, K
%         P1: pressure of state, kPa
%         P0: pressure of environment, kPa
%         v: velocity, m/s^2
%         z: elevation above datum, m
% output: specific exergy (kJ/kg)

result = int_CpAir_dT(T0,T1) -...
      T0*(int_CpAir_divT_dT(T0,T1)-Rair*log(P1/P0)) + v^2/2000 + 9.81*z/1000;

```

```

function result = psi_cg(y,Rcg,T0,T1,P0,P1,v,z)
% Calculates exergy rate at specified state for combustion gas
% Combustion gas components are assumed to be: CO2, H2O, O2, N2
% inputs: y: row vector (y) of # moles of the gas components listed above
%         Rcg:gas constant for combustion gas, kJ/Kg·K
%         T1: temperature of state, K

```

```

%      T0: temperature of environment, K
%      P1: pressure of state, kPa
%      P0: pressure of environment, kPa
%      v:  velocity, m/s^2
%      z:  elevation above datum, m
% output: specific exergy (kJ/kg)

result = int_CpCG_dT(T0,T1,y) -...
        T0*(int_CpCG_divT_dT(T0,T1,y)-Rcg*log(P1/P0)) + v^2/2000 + 9.81*z/1000;

```

```

function R = Rcombgas(y)
% Calculates gas constant (R) for combustion gas
% assumes components are: CO2, H2O, O2, N2
% input: row vector (y) of # moles of the gas components listed above
% output: R (kJ/kgK)

% molar mass of components
M=[44.01 18.015 31.999 28.013];
% calculate molar mass of gas mixture
Mcg = sum(y.*M)/sum(y);
% calculate R
R = 8.314 / Mcg;

```

Appendix VII

Case Study Publication